

The Adjacency Spectrum of the Toroidal Three-Dimensional Queen Graph

Mahesh Ramani

17 August 2026

Abstract

Let G_n be the toroidal three-dimensional queen graph on $(\mathbb{Z}_n)^3$, where n is odd and $3 \nmid n$. Fourier characters diagonalize its adjacency matrix. For $a \in (\mathbb{Z}_n)^3$, let $\mu(a)$ be the number of queen directions orthogonal to a modulo n . The corresponding eigenvalue is

$$\lambda(a) = n\mu(a) - 13.$$

The possible orthogonality counts are exactly

$$\mu(a) \in \{0, 1, 2, 3, 4, 13\}.$$

Writing $M_j(n) = |\{a : \mu(a) = j\}|$, their multiplicities are

$$\begin{aligned} M_{13}(n) &= 1, & M_4(n) &= 9(n-1), & M_3(n) &= 4(n-1), \\ M_2(n) &= 12(n-1), & M_1(n) &= 13n^2 - 72n + 59, & M_0(n) &= n^3 - 13n^2 + 47n - 35. \end{aligned}$$

The proof classifies the frequency points lying on at least two of the thirteen orthogonality hyperplanes and then uses two global counting identities.

1 Introduction

The toroidal three-dimensional queen graph is the periodic analogue of the queen graph on a finite cube. Translation invariance makes it a Cayley graph on $(\mathbb{Z}_n)^3$, so its adjacency operator is diagonal in the Fourier basis. General background on abelian Cayley spectra and finite Fourier analysis may be found in [2, 3]; related two-dimensional queen graphs are treated in [1, 4].

For odd n not divisible by 3, the thirteen toroidal queen directions are distinct. The Fourier eigenvalue at frequency a is determined by the number $\mu(a)$ of direction vectors orthogonal to a . Classifying the intersections of the corresponding thirteen hyperplanes reduces the spectrum to six eigenvalue families. Their multiplicities are explicit polynomials in n .

2 The toroidal queen graph

Definition 2.1. Let $n \geq 1$, and choose one vector from each opposite pair in $\{-1, 0, 1\}^3 \setminus \{0\}$:

$$\begin{aligned} U = \{ & (1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 0), (1, -1, 0), (1, 0, 1), (1, 0, -1), \\ & (0, 1, 1), (0, 1, -1), (1, 1, 1), (1, 1, -1), (1, -1, 1), (1, -1, -1) \}. \end{aligned}$$

The *toroidal three-dimensional queen graph* G_n has vertex set $(\mathbb{Z}_n)^3$. Distinct vertices x, y are adjacent if

$$y = x + tu$$

for some $u \in U$ and $1 \leq t \leq n-1$, with arithmetic modulo n .

Equivalently,

$$G_n = \text{Cay}((\mathbb{Z}_n)^3, S), \quad S = \{tu \bmod n : u \in U, 1 \leq t \leq n-1\}.$$

The set S is closed under negation. Hence G_n is undirected. From now on, assume that

$$n \geq 5, \quad n \text{ is odd}, \quad 3 \nmid n.$$

In this regime the thirteen nonzero cyclic direction sets are disjoint, so G_n is a simple $13(n-1)$ -regular graph.

3 Fourier diagonalization

For $a \in (\mathbb{Z}_n)^3$, define

$$\mu(a) = |\{u \in U : a \cdot u \equiv 0 \pmod{n}\}|.$$

Let $\omega = e^{2\pi i/n}$ and

$$\chi_a(x) = \omega^{a \cdot x}, \quad x \in (\mathbb{Z}_n)^3.$$

Theorem 3.1 (Eigenvalue formula). *For every $a \in (\mathbb{Z}_n)^3$, the character χ_a is an eigenvector of the adjacency matrix of G_n , with eigenvalue*

$$\lambda(a) = n\mu(a) - 13.$$

Proof. Characters diagonalize every Cayley operator on an abelian group. For $x \in (\mathbb{Z}_n)^3$,

$$\begin{aligned} (A\chi_a)(x) &= \sum_{u \in U} \sum_{t=1}^{n-1} \chi_a(x + tu) \\ &= \chi_a(x) \sum_{u \in U} \sum_{t=1}^{n-1} \omega^{t(a \cdot u)}. \end{aligned}$$

For fixed u , the inner sum is $n-1$ when $a \cdot u \equiv 0 \pmod{n}$, and -1 otherwise. Therefore

$$(A\chi_a)(x) = \chi_a(x) (\mu(a)(n-1) - (13 - \mu(a))) = (n\mu(a) - 13)\chi_a(x).$$

□

Corollary 3.2. *The spectrum of G_n is contained in*

$$\{-13, -13 + n, -13 + 2n, \dots, -13 + 13n\}.$$

Moreover, $\lambda(0) = 13(n-1)$, the degree of G_n .

Proof. Theorem 3.1 gives $\lambda(a) = n\mu(a) - 13$ with $0 \leq \mu(a) \leq 13$. At $a = 0$, all thirteen orthogonality relations hold. □

Corollary 3.3. *If $n > 13$, then the minimum adjacency eigenvalue is -13 .*

Proof. For each $u \in U$, the congruence $a \cdot u \equiv 0 \pmod{n}$ defines a subgroup of $(\mathbb{Z}_n)^3$ of size n^2 . Their union has size at most $13n^2 < n^3$. Hence some a satisfies $\mu(a) = 0$, while Theorem 3.1 excludes any eigenvalue below -13 . □

4 Frequency lines

For $u \in U$, put

$$H_u = \{a \in (\mathbb{Z}_n)^3 : a \cdot u \equiv 0 \pmod{n}\}.$$

Thus $\mu(a)$ is the number of hyperplanes H_u containing a .

Lemma 4.1 (Frequency-line classification). *If $a \neq 0$ and $\mu(a) \geq 2$, then, up to a signed coordinate permutation, a lies in one of the cyclic submodules generated by*

$$(1, 0, 0), \quad (1, 1, 0), \quad (1, 1, 1), \quad (1, 1, 2).$$

The nonzero points with $\mu(a) \geq 2$ lie on 25 lines: three axis lines, six face-diagonal lines, four space-diagonal lines, and twelve skew lines.

Proof. Signed coordinate permutations act on the set of orthogonality hyperplanes. It therefore suffices to classify unordered pairs $\{H_u, H_v\}$ up to this action and interchange of u, v . The 78 pairs have the 14 representatives below. Solving the two linear congruences gives the kernel generator in the second column.

$\{u, v\}$	Generator of $H_u \cap H_v$
$\{(1, 1, 1), (1, 1, 0)\}$	$(1, -1, 0)$
$\{(1, 1, 1), (1, 1, -1)\}$	$(1, -1, 0)$
$\{(1, 1, 1), (1, 0, 0)\}$	$(0, 1, -1)$
$\{(1, 1, 1), (1, 0, -1)\}$	$(1, -2, 1)$
$\{(1, 1, 1), (1, -1, -1)\}$	$(0, 1, -1)$
$\{(1, 1, 1), (0, 0, -1)\}$	$(1, -1, 0)$
$\{(1, 1, 1), (0, -1, -1)\}$	$(0, 1, -1)$
$\{(1, 1, 0), (1, 0, 1)\}$	$(1, -1, -1)$
$\{(1, 1, 0), (1, 0, 0)\}$	$(0, 0, 1)$
$\{(1, 1, 0), (1, -1, 0)\}$	$(0, 0, 1)$
$\{(1, 1, 0), (0, 0, 1)\}$	$(1, -1, 0)$
$\{(1, 1, 0), (0, -1, 1)\}$	$(1, -1, -1)$
$\{(1, 1, 0), (0, -1, 0)\}$	$(0, 0, 1)$
$\{(1, 0, 0), (0, 1, 0)\}$	$(0, 0, 1)$

The resulting kernels belong to the four stated signed-permutation families. These families contain, respectively, 3, 6, 4, and 12 lines. Their coordinate patterns distinguish them, and the assumptions that 2 and 3 are units modulo n keep the listed lines distinct. Each generator has a unit coordinate, so each line contains exactly n points. This accounts for every nonzero frequency with $\mu(a) \geq 2$. $\square$

Theorem 4.2 (Exact multiplicities). *The possible values of $\mu(a)$ are*

$$\{0, 1, 2, 3, 4, 13\}.$$

If

$$M_j(n) = |\{a \in (\mathbb{Z}_n)^3 : \mu(a) = j\}|,$$

then

$$M_{13}(n) = 1, \quad M_4(n) = 9(n-1), \quad M_3(n) = 4(n-1), \quad M_2(n) = 12(n-1),$$

and

$$M_1(n) = 13n^2 - 72n + 59, \quad M_0(n) = n^3 - 13n^2 + 47n - 35.$$

Equivalently, the adjacency eigenvalues are

$$\lambda_j = jn - 13, \quad j \in \{0, 1, 2, 3, 4, 13\},$$

with the corresponding multiplicities above. For $n = 5$ and $n = 7$, $M_0(n) = 0$; for every admissible $n > 7$, all six multiplicities are positive.

Proof. By Lemma 4.1, every nonzero frequency with $\mu(a) \geq 2$ lies on one of the 25 listed lines. Direct substitution shows that the three axis and six face-diagonal lines have $\mu = 4$, the four space-diagonal lines have $\mu = 3$, and the twelve skew lines have $\mu = 2$. Each line has $n - 1$ nonzero points. Thus

$$M_4(n) = 9(n - 1), \quad M_3(n) = 4(n - 1), \quad M_2(n) = 12(n - 1).$$

The origin lies on all thirteen hyperplanes, so $M_{13}(n) = 1$.

The total number of frequencies gives

$$M_1(n) + M_0(n) = n^3 - (1 + 25(n - 1)) = n^3 - 25n + 24.$$

Also,

$$\sum_{a \in (\mathbb{Z}_n)^3} \mu(a) = 13n^2,$$

because each H_u contains n^2 points. Substituting the multiplicities already determined yields

$$13 + 4 \cdot 9(n - 1) + 3 \cdot 4(n - 1) + 2 \cdot 12(n - 1) + M_1(n) = 13n^2.$$

Hence

$$M_1(n) = 13n^2 - 72n + 59,$$

and subtraction from the total count gives

$$M_0(n) = n^3 - 13n^2 + 47n - 35.$$

□

5 Example

Example 5.1 (The case $n = 5$). Here $M_0(5) = 0$, so -13 is absent. The spectrum is

μ	$\lambda = 5\mu - 13$	Multiplicity
13	52	1
4	7	36
3	2	16
2	-3	48
1	-8	24

The multiplicities sum to 125, and the eigenvalue sum is

$$52 + 36 \cdot 7 + 16 \cdot 2 + 48(-3) + 24(-8) = 0,$$

as required for an adjacency matrix with zero diagonal.

6 Conclusion

In the generic odd regime, the toroidal three-dimensional queen graph has six eigenvalue families. Their multiplicities follow from the 25 frequency lines on which at least two orthogonality conditions hold, together with the total-frequency and total-incidence counts.

When n is even or divisible by 3, direction collisions and additional modular intersections occur. Those arithmetic cases require a separate frequency-line classification.

References

- [1] D. M. Cardoso, I. S. Costa, and R. Duarte, Spectral properties of the n -queens' graphs, arXiv:2012.01992 (2020).
- [2] C. Godsil and G. Royle, *Algebraic Graph Theory*, Graduate Texts in Mathematics 207, Springer, 2001.
- [3] A. Terras, *Fourier Analysis on Finite Groups and Applications*, London Mathematical Society Student Texts 43, Cambridge University Press, 1999.
- [4] W. D. Weakley, The automorphism group of the toroidal queen's graph, *Australasian Journal of Combinatorics* 42 (2008), 141–158.